Q1 - Single Correct

The point $P(x, y)$ moves in XY -plane in such a way that $[|x|]+[|y|]=1$, where $[\cdot]$ denotes the greatest integer function. Area of the region representing all possible positions of the point P, is

(1) 4 sq units

(2) 16 sq units

(3) $2\sqrt{2}$ sq units

(4) 8 sq units

Q2 - Single Correct

The area bounded by the minimum $\{|x|,|y|\}=1$ and the maximum of $\{||x|,|y||\}$ =2 is

(1) 4 sq units

(2) 8 sq units

(3) 16 sq units

(4) 9 sq units

Q3 - Single Correct

A square $ABCD$ is inscribed in a circle of radius 4 and a point P moves inside the circle such that $d(P, AB) \leq \min\{d(P, BC), d(P, DA)\}$, where $d(P, AB)$ is the distance of a point P from the line AB. The area of region (in sq units) covered by the moving point P is

(1) 4π

(2) 8π

(3) $8\pi-16$

(4) $4\pi-4$

Q4 - Single Correct

The area of the loop of the curve $ay^2=x^2(a-x)$ is

(1) $4a^2$ sq units

(2) $\frac{8a^2}{15}$ sq units

(3) $\frac{16a^2}{9}$ sq units

(4) None of these

Q5 - Single Correct

Three points $O(0,0), P\left(a, a^2\right)$ and $Q\left(-b, b^2\right)(a>0, b>0)$ are on the parabola $y=x^2$. Let S_1 be the area bounded by the line PQ and the parabola and let S_2 be the area of the ΔOPQ, then the minimum value of S_1/S_2, is

(1) $\frac{4}{3}$

(2) 5

(3) 2

(4) $\frac{7}{3}$

Q6 - Single Correct

$y=f(x)$ is a function which satisfy

(i) $f(0)=0$

(ii) $f''(x)=f'(x)$

(iii) $f'(0)=1$

Then, the area (in sq units) bounded by the graph of $y=f(x)$, the lines $x=0, x-1=0$ and $y+1=0$, is

(1) e

(2) $e-2$

(3) $e-1$

(4) $e+1$

Q7 - Multiple Correct

Let T be the triangle with vertices $(0,0),\left(0, c^2\right),\left(c, c^2\right)$ and R be the region between $y=cx$ and $y=x^2$, where $c>0$, then

(1) area $(R)=\frac{c^{3}}{6}$

(2) area $(R)=\frac{c^{3}}{3}$

(3) $\lim_{c \rightarrow 0^{+}} \frac{\operatorname{area}(T)}{\operatorname{area}(R)}=3$

(4) $\lim_{c \rightarrow 0^{+}} \frac{\operatorname{area}(T)}{\operatorname{area}(R)}=\frac{3}{2}$

Q8 - Multiple Correct

Suppose f is defined from $R \rightarrow[-1,1]$ as $f(x)=\frac{x^{2}-1}{x^{2}+1}$, where R is the set of real numbers. Then, which statements does not hold?

(1) f is many-one and onto

(2) f increases for $x>0$ and decreases for $x<0$

(3) Minimum value is not attained even though f is bounded

(4) The area included by the curve $y=f(x)$ and the line $y=1$ is π sq units

Q9 - Multiple Correct

Consider $f(x)=\begin{cases}\cos x, & 0 \leq x<\frac{\pi}{2} \\ \left(\frac{\pi}{2}-x\right)^{2}, & \frac{\pi}{2} \leq x<\pi\end{cases}$, such that f is periodic with period π, then

(1) the range of f is $\left[0, \pi^{2}/4\right]$

(2) f is continuous for all real x, but not differentiable for some real x

(3) f is continuous for all real x

(4) the area bounded by $y=f(x)$ and the X -axis from $x=-n\pi$ to $x=n\pi$ is $2n\left(1+\frac{\pi^{3}}{24}\right)$ for a given $n \in N$.

Q10 - Multiple Correct

Consider the functions $f(x)$ and $g(x)$, both defined from $R \rightarrow R$ and are defined as $f(x)=2x-x^{2}$ and $g(x)=x^{n}$, where $n \in N$. If the area between $f(x)$ and $g(x)$ is $1/2$, then n is a divisor of

(1) 12

(2) 15

(3) 20

(4) 30

Q11 - Multiple Correct

If A_i is the area bounded by $|x-a_i|+|y|=b_i, i \in N$, where $a_{i+1}=a_i+\frac{3}{2}b_i$ and $b_{i+1}=\frac{b_i}{2}$, $a_1=0, b_1=32$, then

(1) $A_3=128$

(2) $A_3=256$

(3) $\lim_{n\to\infty}\sum_{i=1}^{n} A_i=\frac{8}{3}(32)^2$

(4) $\lim_{n\to\infty}\sum_{i=1}^{n} A_i=\frac{4}{3}(16)^2$

Q12 - Paragraph 1

Passage I (For Question 12, 13)

Let $f(x)=x^2-3x+2$ be a function, $\forall x \in R$.

The area bounded by the curve $f(|x|)$ and X -axis, is

(1) 2 sq units

(2) $5/2$ sq units

(3) $3/5$ sq unit

(4) $5/3$ sq units

Q13 - Paragraph 1

The area bounded by the curve $|f(|x|)|$ between $1 \leq |x| \leq 2$ and the X -axis is

(1) $1/5$ sq unit

(2) $1/4$ sq unit

(3) $1/3$ sq unit

(4) $1/2$sq unit

Q14 - Integer Type

If A is the area between the curve $y=x^{6}(1-x)^{7}$ and X -axis, then the number of prime factors of A^{-1} is

Q15 - Integer Type

Area of the figure contained between the parabola $x^{2}=4y$ and the curve $y=\frac{8}{x^{2}+4}$ is $\left(4\pi-\frac{k}{3}\right)$, then the value of k is

Q16 - Integer Type

Let $f(x)=\text{maximum}\left\{x^{2},(1-x)^{2}, 2x(1-x)\right\}$, where $0 \leq x \leq 1$. If the area of the region bounded by the curves $y=f(x)$, x -axis, $x=0$ and $x=1$ is $\left(\frac{m}{n}\right)$, then $(2m-n)$ is equal to

Answer Key

Q1 (4)	**Q2** (1)	**Q3** (1)	**Q4** (2)
Q5 (1)	**Q6** (3)	**Q7** (1, 3)	**Q8** (1, 3, 4)
Q9 (1, 4)	**Q10** (2, 3, 4)	**Q11** (1, 3)	**Q12** (1)
Q13 (3)	**Q14** (5)	**Q15** (4)	**Q16** (7)